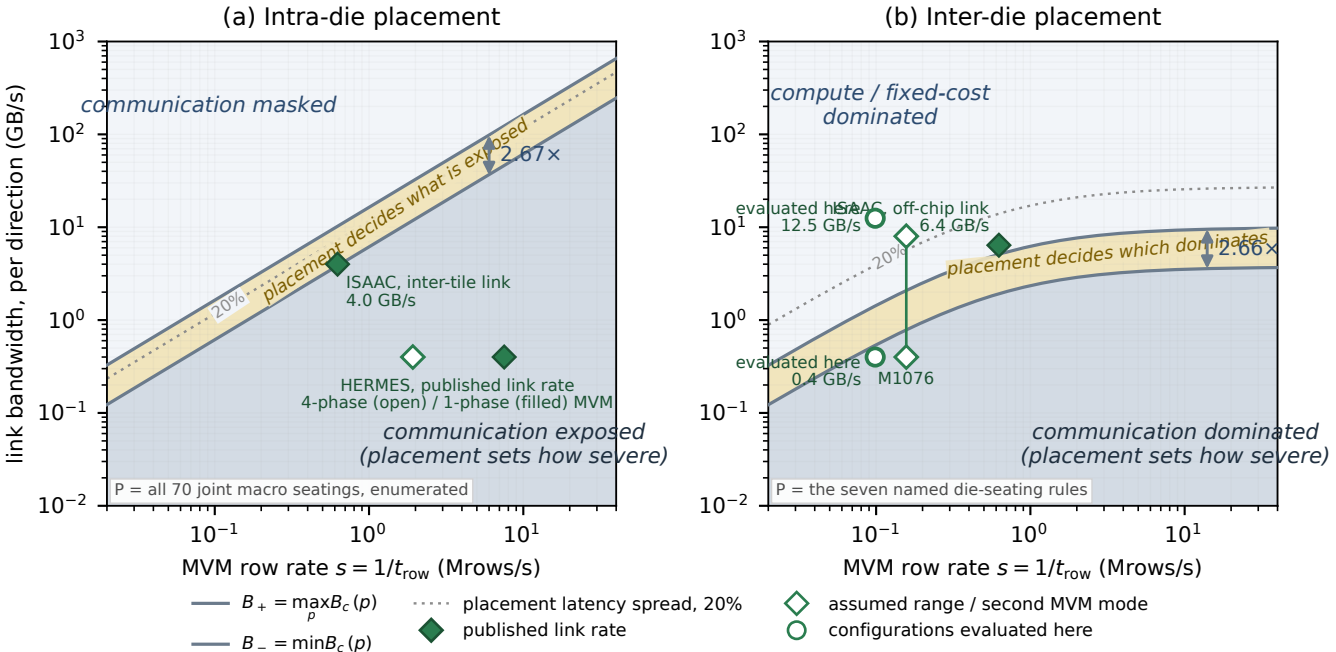
(a) Intra-die placement
(b) Inter-die placement
link bandwidth, per direction (GB/s)
MVM row rate $s = 1/t_{row}$ (Mrows/s)
communication masked
placement decides what is exposed
2.67×
20%
ISAAC, inter-tile link
4.0 GB/s
HERMES, published link rate
4-phase (open) / 1-phase (filled) MVM
communication exposed
(placement sets how severe)
P = all 70 joint macro seatings, enumerated
compute / fixed-cost
dominated
evaluated here
12.5 GB/s
ISAAC, off-chip link
6.4 GB/s
placement decides which dominates
2.66×
evaluated here
0.4 GB/s
M1076
communication dominated
(placement sets how severe)
P = the seven named die-seating rules
$B_+ = \max_p B_c(p)$
$B_- = \min_p B_c(p)$
placement latency spread, 20%
published link rate
assumed range / second MVM mode
configurations evaluated here